

1 Solution — GIAM 7.1

Problem 6

1.1 Problem

How many “words” are there of length 4, with distinct letters from the Cryptographer’s alphabet, in which the letters appear in *increasing* order alphabetically? (“**acef**” would be one such word, but “**cafe**” would not.)

1.2 Setup

The alphabet is $\Sigma = \{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{e}, \mathbf{f}\}$ carrying its usual total order $\mathbf{a} < \mathbf{b} < \mathbf{c} < \mathbf{d} < \mathbf{e} < \mathbf{f}$.

A *valid word* (for this problem) is a length-4 tuple $(x_1, x_2, x_3, x_4) \in \Sigma^4$ satisfying

$$x_1 < x_2 < x_3 < x_4.$$

Note that the strict inequality automatically forces the four letters to be *distinct*: $x_i = x_j$ is impossible whenever $x_i < x_j$.

1.3 Count — Subsets in Disguise

Define the map

$$\Phi : \{\text{valid words}\} \longrightarrow \{4\text{-element subsets of } \Sigma\}, \quad \Phi(x_1, x_2, x_3, x_4) = \{x_1, x_2, x_3, x_4\}.$$

This Φ is a *bijection*:

- **Well-defined.** A valid word has four distinct entries, so its image is a 4-subset.
- **Surjective.** Every 4-subset $A \subseteq \Sigma$ has exactly one increasing-order arrangement — list its elements in alphabetical order — and that arrangement is a valid word with Φ -image A .
- **Injective.** Two valid words with the same underlying set must each list its elements in the unique increasing order, hence be equal.

So valid words of length 4 correspond one-to-one with 4-element subsets of Σ , and

$$\# \text{valid words} = \binom{6}{4} = \binom{6}{2} = \frac{6 \cdot 5}{2} = \boxed{15}.$$

(The middle equality uses the symmetry $\binom{n}{k} = \binom{n}{n-k}$ — exactly the identity proved in Problem 4 of this section.)

1.4 Enumeration

Grouping the 15 valid words by their first letter x_1 gives a clean tally. Once x_1 is fixed, the remaining three letters form a 3-subset of the alphabet entries strictly greater than x_1 , arranged in alphabetical order:

First letter x_1	$\{x_2, x_3, x_4\}$ is a 3- subset of	Count	The words
a	$\{b, c, d, e, f\}$	$\binom{5}{3} = 10$	abcd, abce, abcf, abde, abdf, abef, acde, acdf, acef, ade
b	$\{c, d, e, f\}$	$\binom{4}{3} = 4$	bcde, bcdf, bcef, bdef
c	$\{d, e, f\}$	$\binom{3}{3} = 1$	cdef
d, e, f	(fewer than 3 larger letters remain)	0	—
Total	—	10 + 4 + 1 = 15	—

Table 1.1

This first-letter decomposition is an instance of the **hockey-stick identity**,

$$\binom{6}{4} = \binom{5}{3} + \binom{4}{3} + \binom{3}{3} = 10 + 4 + 1 = 15,$$

where each summand $\binom{m}{3}$ counts the valid words whose first letter is the unique element of Σ with m strictly-larger neighbors.

1.5 Remark — One Increasing Arrangement Per Subset

It is instructive to relate this count to two neighbors in the problem-design hierarchy:

Constraint on the length-4 word	Count
any letters, any order (Problem 5)	$6^4 = 1,296$
<i>distinct</i> letters, <i>any</i> order	$6 \cdot 5 \cdot 4 \cdot 3 = 360$
<i>distinct</i> letters, <i>increasing</i> order (this problem)	$\binom{6}{4} = 15$

Table 1.2

The third row arises from the second by the observation that each 4-subset $A \subseteq \Sigma$ admits $4! = 24$ orderings, exactly one of which is increasing. Hence the **360** distinct-letter ordered words partition into groups of **24** by their underlying subset, and

$$\# \text{increasing words} = \frac{360}{4!} = \frac{360}{24} = 15 = \binom{6}{4},$$

recovering the standard identity $\binom{n}{k} = \frac{P(n, k)}{k!}$ (the number of k -permutations divided by $k!$).